

Designing Celebratory Technology for Neurodiversity with Neurodivergent Scholars

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ABSTRACT

Research indicates that the social stigma of an autism label is the biggest unaddressed problem with autism. To address this gap, a novel celebratory paradigm will seed the first wave of Celebratory Technology for Neurodiversity. By creating Celebratory Technology SIG, we plan to foster an HCI community of neurodiverse scholars over the coming years. A goal of the SIG is to enrich HCI researchers' understanding of neurodiversity because of increased awareness of neurodiverse communities, approach design from a celebratory perspective that aims to intervene at a societal level, and clarity around community-oriented challenges that warrant further exploration. This SIG will produce knowledge via a conceptual framework and empirically derived interaction designs for social change. Design sprint results and discussion will be encouraged to be expanded upon and disseminated via publications in venues such as ACM CHI, ASSETS, and TACCESS. Ultimately, these new interaction designs may drive programs implemented by social media platforms and policy to reduce autism stigma.

CCS CONCEPTS

• **Interaction Design**; • **Accessibility**; • **Social Justice**;

KEYWORDS

Neurodiversity, Celebratory Technology

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1 INTRODUCTION

Research indicates that the social stigma of an autism label is the biggest unaddressed problem with autism [3]. This SIG aims to explore a new technology paradigm to reduce the social stigma of autism. The autistic experience of stigma has recently become a focus of study yet technological interventions have not kept pace. One study found that of teens who admitted social difficulties, almost 80% perceived stigma from their autism label [1]. Furthermore,

within our own field of computer science, some studies have found male college students and STEM majors report higher levels of stigma compared to Humanities majors [4]. However, mechanisms to reduce stigma can be engineered into existing sociotechnical systems such as social media applications and groupware systems such as Zoom. In this SIG meeting, we will design for Neurodiversity and offer a new paradigm—Celebratory Technologies for Neurodiversity [2].

Celebratory Technology for Neurodiversity strives to combat social stigma on a broad scale. Instead of perpetuating a narrow narrative that portrays disability as inherently lesser than, with sporadic instances of exceptional or ordinary inspiration, Celebratory Technologies actively promotes a positive and inclusive story that encompasses the diverse experiences of individuals. The emphasis is on acknowledging neurodiversity as an integral aspect of our shared reality, refraining from isolating neurodivergence as an exceptional use case. This innovative paradigm, Celebratory Technology for Neurodiversity, introduces a fundamental shift. Rather than focusing on adapting the skillsets of neurodivergent individuals or accommodating them within existing societal frameworks, it addresses the root cause—society's pervasive biases. By doing so, it acknowledges that the responsibility for change lies not solely with the neurodivergent individual but with society at large. This paradigm shift goes beyond the confines of traditional research, extending its scope beyond the individual with autism and those immediately connected to them. It creates a space for design that transcends the experiences of a select few, calling upon everyone to embrace the collective humanity that unites us.

2 BACKGROUND ON CELEBRATORY TECHNOLOGY FOR NEURODIVERSITY

Celebratory Technology for Neurodiversity is an emerging paradigm of interaction design that targets the public's perception of neurodiversity. The initial goals of Celebratory Technology are to raise the positive aspects of neurodivergence as well as provide awareness of neurodivergent perspective; acceptance of neurodiversity as a natural difference and not less than neurotypical; insightful accommodations neurodivergence thus providing access to resources, culture, relationships, and work opportunities; advocacy for equity of resource distribution; and appreciation for natural spectrum of human expression. To crystallize this concept into a paradigm, international engagement is required.

The first step of crystallizing celebratory Technology for Neurodiversity began at the "Celebrating Neurodiversity: Ichi-go ichi-e Symposium" held in Tokyo, Japan, in September 2023. The authors organized this event with support from Chiba Institute of Technology, Digital Garage, and the Connected Learning Alliance in

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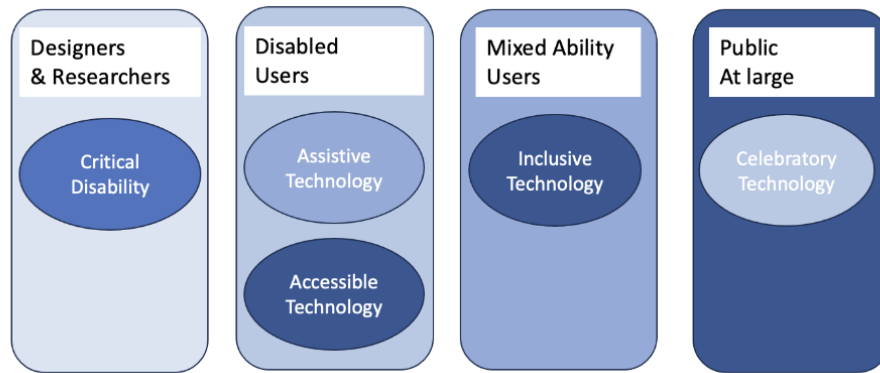


Figure 1: Neurodiversity Ecosystem Based on Use Case and Target Audience.

collaboration with Joi Ito and Mimi Ito. The symposium brought together diverse perspectives from stakeholders and thought leaders in academic research and autism advocacy. A group of twenty international human-computer interaction (HCI) scholars and autism advocates engaged in three days of compelling discussions, insightful presentations, and rich cultural explorations in Tokyo, and a collaborative design sprint. The symposium also featured an afternoon public event with simultaneous translation in Japanese, focusing on neurodiversity as a driving social movement and a promising research domain. The Neurodiversity Connected Learning Symposium in Tokyo 2023 enriched the group's understanding of neurodiversity and also inspired actionable steps toward a more inclusive and accommodating society. Evolving Celebratory Technology and bringing it to academic audiences is one such actionable step.

This SIG will continue exploration through the academic forum. Just as the aforementioned symposium encouraged cross-discipline collaboration, actively worked to bridge the gap in the perceived clash of epistomes in the field of neurodiversity, and promoted a framework to incorporate each perspective, this SIG will adhere to these principles through a design sprint grounded in a shared vocabulary for the ecosystem of technologies for neurodiversity. As depicted in Figure 1, four related paradigms lay out an ecosystem to consider technology across a wide frame of support. Each has a distinct primary user group. The assistive technology use case targets skills deficits conceptualized as differences within the neurodivergent users. The accessible technology use case aims to work around challenges (presumed to be socially constructed) to access information. The inclusive technology use case aims to broaden the user of a system beyond mainstream users. Critical designs aim to address the gap between wants and needs of disabled communities from the often deficit-oriented “solutions” created by scholars targeting neurodivergence. The new use case of celebratory technology for neurodiversity aims to create a positive projection of the spectrum of human diversity to the broader public to reduce stigma of neurodivergence. The SIG will utilize the design sprint approach of the symposium, providing an opportunity to share insights from the symposium with the SIG to seed novel design concepts.

3 METHODS

Through a collaborative design sprint facilitated by the organizers, we will explore research directions with neurodiversity community-aligned end users goals including consciously addressing culturally sensitive social norms with an individual's right to agency, authenticity, play, and joy. We will pioneer emerging Celebratory Technology interaction designs to promote the strengths of neurodiversity. This SIG takes a design approach to raising awareness of neurodiversity, engaging with neurodiversity-related data, and generating design concepts that foreground people's strengths and interests. We will conduct a design sprint at the SIG meeting in the spirit of a designathon [5].

4 GOALS FOR THIS SPECIAL INTEREST GROUP (SIG) MEETING

The goals for this SIG include (1) advancing the emerging conceptual framing of Celebratory Technology for Neurodiversity; (2) foregrounding neurodivergent strengths and building collaborative capacities in meaningful ways, and (3) collaboratively exploring methods for societal inclusion in the HCI research domain. This SIG will build an inclusive community of CHI researchers who have neurodivergent and neurotypical lived experiences. Additionally, this SIG will broaden the reach of neurodiversity research within CHI by providing an on-ramp and entry point to engaging with the neurodiverse socio-technical landscape.

Brief overview of the neurodiversity paradigm and description of Celebratory Technology.

SIG ice breaker to facilitate small group introductions

Conduct Design Sprint:

Step 1: Empathize with local and global neurodivergent communities through neurodiversity resources, which may include written vignettes, short videos, and local neurodiversity community-based resources. These resources will be collected by SIG organizers prior to session

Step 2: Define design space based on participant use cases as much as possible

Step 3: Rapid ideation in small groups

Step 4: Develop lo fi prototype (e.g., paper prototype)

Step 5: Lightning presentation of concept to group and receive brief feedback

Next steps Discussion (i.e., coordinate a slack channel for ongoing collaboration).

Follow Reflection via Google Docs post session.

ACKNOWLEDGMENTS

We would like to thank the participants and other organizers of the Japan symposium.

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A APPENDIX: SIG SUPPLEMENTARY MATERIAL

1. A brief description of the community (or communities) to which this SIG would be of interest and why it is of interest:

HCI researchers are interested in the Neurodiversity movement for professional and personal reasons as someone who identifies as neurodivergent or is an ally to the community.

Assumed attendee background: We welcome anyone with an interest to collaborate from the novice to the expert who is very knowledgeable in the debates that occur in this space around episodic view of autism versus neurodivergent.

The approach you will use for organizing and presenting the SIG: We will have an opening and closing activity with a Celebratory Technology design sprint as the main activity.

0. informal schedule of discussion topics

This agenda provides a structured approach to exploring neurodiversity in design within a 75-minute timeframe, ensuring each activity receives adequate attention and participation.

Detailed Agenda:

- Introduction (5 minutes)

o Welcome participants to the meeting.

o Brief overview of the meeting's purpose and structure.

- Overview of the Neurodiversity Paradigm (5 minutes)

o Introduction to the neurodiversity paradigm.

o Brief overview of autism from the perspective of neurodiversity.

o Explanation of Celebratory Technology and its significance.

- SIG Ice Breaker (5 minutes)

o Ice breaker activity to facilitate small group introductions.

o Break the ice and create a comfortable environment for collaboration.

- Design Sprint

- Step 1: Empathize (10 minutes)

o Engage with neurodiversity resources sourced by SIG organizers.

o Utilize written vignettes, short videos, and neurodiversity community-based resources.

Step 2: Define (10 minutes)

- Define the design space based on SIG-sourced use cases as much as possible.

Step 3: Ideate (15 minutes)

- Rapid ideation in small groups.

Step 4: Prototype (15 minutes)

- Develop analog low-fidelity prototype (e.g., interface sketch).

Step 5: Presentation and Feedback (10 minutes)

- Lightning presentation of concepts to the group.

- Receive brief feedback from participants.

- Follow-Up Reflection on SIG and Neurodiversity at CHI (5 minutes)

o Use Google Docs for post-session reflection.

o Encourage participants to share insights and thoughts after the meeting.

- Next Steps and Closing (5 minutes)

o Discuss further actions and how to stay connected beyond the meeting.

o Thank participants for their engagement and contributions.

o Highlight the importance of continued collaboration in neurodiversity-inclusive design.

o Coordinate a Slack channel for ongoing collaboration. Share the link to the Slack channel.

0. Plan to attract a reasonable number of attendees.

We plan to use social media to promote the Celebratory Technology for Neurodiversity SIG as well as directly email and slack the cohort of people we each know in this space, and we will ask them to pass it to their network as well.

1. **LouAnne Boyd will serve as the primary organizer and should serve as the primary contact.**